



TOPICAL REVIEW

Progression of total training volume in resistance training studies and its application to skeletal muscle growth

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E-mail: jploenne@olemiss.edu**Keywords:** progressive overload, muscle growth, resistance training volume**Abstract**

Progressive overload describes the gradual increase of stress placed on the body during exercise training, and is often quantified (i.e. in resistance training studies) through increases in total training volume (i.e. sets $\times$ repetitions $\times$ load) from the first to final week of the exercise training intervention. Within the literature, it has become increasingly common for authors to discuss skeletal muscle growth adaptations in the context of increases in total training volume (i.e. the magnitude progression in total training volume). The present manuscript discusses a physiological rationale for progressive overload and then explains why, in our opinion, quantifying the progression of total training volume within research investigations tells very little about muscle growth adaptations to resistance training. Our opinion is based on the following research findings: (1) a noncausal connection between increases in total training volume (i.e. progressively overloading the resistance exercise stimulus) and increases in skeletal muscle size; (2) similar changes in total training volume may not always produce similar increases in muscle size; and (3) the ability to exercise more and consequently amass larger increases in total training volume may not inherently produce more skeletal muscle growth. The methodology of quantifying changes in total training volume may therefore provide a means through which researchers can mathematically determine the total amount of external ‘work’ performed within a resistance training study. It may not, however, always explain muscle growth adaptations.

1. Introduction

Progressive overload describes the gradual increase of stress placed on the body during exercise training. This concept was introduced as early as 1945 by DeLorme (1945) and now serves as a fundamental principle within resistance training guidelines (Ratamess *et al* 2009, Garber *et al* 2011, Haff and Triplett 2016). The American College of Sports Medicine (Ratamess *et al* 2009) recommends that there must be an increase in the demands placed on the musculature to allow the possibility for increases in skeletal muscle size. In other words, as individuals adapt to resistance exercise (i.e. become more ‘resistance trained’), it is believed that a larger stimulus is required to elicit additional muscle growth. Within the resistance training literature, it is common for researchers to calculate total training volume (i.e. sets $\times$ repetitions $\times$ load) as a means to estimate the dose of the exercise stimulus imposed on the working musculature (Nunes *et al* 2021). Recent studies (Barcelos *et al* 2018, Damas *et al* 2019, Corrêa *et al* 2021, Brigatto *et al* 2022, Krause Neto *et al* 2022, Nóbrega *et al* 2023, Kassiano *et al* 2023) have employed similar methods, and quantified changes in total training volume from the initial to final training weeks (i.e. the ‘progression’ of total training volume) to better understand muscle growth adaptations. The idea that progression in total training volume within research studies can explain the muscle growth response to resistance training assumes that progressive overload is driving hypertrophic adaptations. The present manuscript discusses such an assumption and

explains why, in our opinion, quantifying the progression of total training volume within research investigations tells very little about muscle growth adaptations.

2. Literature search

Damas *et al* (2019) were the first authors, to our knowledge, to specifically plot increases in total training volume across each week of training as a means to quantify the magnitude of progressive overload for different resistance training protocols (i.e. the 'progression' of total training volume). As such, we originally used Google Scholar to find articles that cited the work of Damas *et al* (2019), and then included additional references where appropriate to help illustrate the topic discussed herein (i.e. not all studies mentioned in the paper specifically cited (Damas *et al* 2019)). We elected to use a narrative format, as opposed to a systematic search of the literature, given that progressive overload is a concept inherent to the large majority, if not the entirety, of the resistance training literature. It is acknowledged that such an approach poses a risk of study selection bias (Garg *et al* 2008), and may be perceived as a limitation to our discussion.

3. A physiological rationale for progressive overload

Muscle growth following resistance training is believed to occur through the process of mechanotransduction (Marcotte *et al* 2015, Lim *et al* 2022), in which a muscle cell senses a mechanical stimulus and activates a network of signaling pathways that shift protein turnover to favor a net positive protein balance (Rennie *et al* 2004). Mechanotransduction can thus be considered a local (i.e. within the muscle) biological process that underpins skeletal muscle growth (i.e. an 'internal variable') (Lim *et al* 2022). Given that the hypertrophic effect appears to be related to the proportion of the muscle activated during training (Marcotte *et al* 2015), employing progressive overload might be understood as an attempt to ensure that the exercise stimulus (i.e. an 'external variable' Lim *et al* 2022) remains sufficient for increasing muscle size (i.e. capable of triggering the mechanotransduction cascade across a large majority of the muscle). Quantifying total training volume (sets $\times$ repetitions $\times$ load) may provide a numerical estimate of the external stimulus imposed on the working musculature (the total amount of 'work' performed i.e. an external variable (Lim *et al* 2022) (Nunes *et al* 2021); however, it is important to remember that the response of the internal (not external) variables is what appears to be the key determinant of skeletal muscle growth (Lim *et al* 2022).

Indirect evidence to support a physiological rationale for progressive overload can be inferred from Ploutz *et al*'s (1994) work. They demonstrated that, after 9 weeks of progressive, unilateral knee extension training, less overall muscle (quadriceps femoris) was required to lift the same submaximal exercise loads, for the same number of repetitions compared to pre-intervention. Ploutz *et al* (1994) attributed these findings to neural adaptations given that the same effect (i.e. less muscle mass being activated to lift the same submaximal loads) was observed in the contralateral untrained limb which experienced increases in muscle strength but not muscle size. The physiological consequence of recruiting less muscle (per repetition) to lift a given absolute load (for the same number of repetitions) would be that fewer muscle fibers receive the mechanotransduction stimulus. Phillips *et al* (2002) reported that an isolated bout of resistance exercise in the non-resistance trained state (i.e. before 8 weeks of progressive resistance training) increased mixed muscle protein synthesis acutely, whereas the protein synthetic response to the same absolute workload was attenuated following 8 weeks of training. Although the greater rise in mixed protein synthesis in the non-resistance trained state could have been due to repair and remodeling (i.e. muscle damage) (Damas *et al* 2018), it should be noted that using the same absolute workload likely required much less effort in the resistance trained state (because of increased maximum strength).

A follow up, within-subject study (Tang *et al* 2008) evaluated individuals at the same relative training load, and found that, at post-intervention, the resistance trained limb performed a larger volume of exercise and observed greater increases in mixed muscle protein synthesis (4 h postexercise) compared to the non-resistance trained limb. Yet, at 28 h postexercise, the trained limb showed an attenuated protein synthetic response (Tang *et al* 2008). Together these findings provide some reason to think that progressive overload would be important to facilitate additional increases in muscle size.

4. Progression of total training volume and its influence on skeletal muscle growth

To allow the possibility for increases in muscle size, it is recommended that, over time, an increase in the demands should be placed on the working musculature (i.e. progressive overload) (Ratamess *et al* 2009, Garber *et al* 2011, Haff and Triplett 2016). Individuals can adhere to the principle of progressive overload in several different ways, for example: (1) increasing the absolute load in a given repetition maximum (RM) zone (e.g. 8–12RM) (Kataoka *et al* 2022, Hammert *et al* 2023b, 2023c); (2) increasing the number of

repetitions at a specific training load (i.e., percentage one-RM; 1RM) (Jessee *et al* 2018, Plotkin *et al* 2022); and/or (3) increasing the number of sets (to failure) per training session and/or week (Haun *et al* 2018, Hammert *et al* 2023a). Undertaking any of these modifications will inevitably result in an increase in total training volume. Perhaps for this reason, researchers (Barcelos *et al* 2018, Damas *et al* 2019, Corrêa *et al* 2021, Brigatto *et al* 2022, Krause Neto *et al* 2022, Nóbrega *et al* 2023, Kassiano *et al* 2023) have begun to use increases in total training volume (i.e. from the initial to final training weeks) as a means to quantify and compare the magnitude of progressive overload between resistance training protocols (i.e. the ‘progression’ of total training volume). Damas *et al* (2019), who examined muscle growth adaptations between a standard progressive training protocol (i.e. 4 sets of 9–12RM) and a variable training protocol that systematically and sequentially (i.e. every session) manipulated different training variables, observed that the progression of total training volume followed a similar pattern between the two training protocols (Damas *et al* 2019). Given that the two protocols also demonstrated similar increases in muscle size, it was hypothesized that the progression of total training volume throughout a resistance training period could be important for maximizing muscle growth adaptations (Damas *et al* 2019). Yet, because Damas *et al* (2019) did not include training conditions that performed the same protocols without implementing progressive overload, it is difficult to know whether the increases in total training volume were truly important for facilitating muscle growth. Nonetheless, published studies (Corrêa *et al* 2021, Brigatto *et al* 2022, Krause Neto *et al* 2022, Nóbrega *et al* 2023, Kassiano *et al* 2023) have cited Damas *et al*’s (2019) work and employed similar methodologies (e.g. reporting increases in total training volume from the initial to final training weeks), presumably in an attempt to better understand muscle growth outcomes.

Using data from two different experiments (Barcelos *et al* 2018, Nóbrega *et al* 2018), Nóbrega *et al* 2023 investigated the effects of different volume load progression models on changes in muscle size. The two experiments had participants perform 24 total sessions of knee extension exercise (Barcelos *et al* 2018, Nóbrega *et al* 2018). Individuals who exercised to failure (3x/week; 8 weeks) and had their training loads adjusted to maintain a 9–12RM (i.e. each set and throughout the training study) reportedly experienced more growth and progression in volume load than those who exercised to failure (2x/week; 12 weeks) at 80% 1RM each set and training session. The differences in muscle growth adaptations was rationalized by the suggestion that the set by set load adjustments ensured that the stimulus remained at an ‘optimal’ range for adaptive capacity (i.e. promoted increases in volume load at an accelerated pace), and was thus more effective increasing muscle size (Nóbrega *et al* 2023). It must be noted, however, that the use of a non-randomized, retrospective analysis limits the ability to draw causal inferences on the topic, and that the two protocols not only differed in terms of progression models, but also the number of training sessions per week (i.e. training frequency) (Barcelos *et al* 2018, Nóbrega *et al* 2018). Two other investigations (Corrêa *et al* 2021, Brigatto *et al* 2022) aimed to determine the influence of training frequency on muscle growth adaptations, both of which were unable to identify greater changes in muscle size higher frequency training. The authors (Corrêa *et al* 2021, Brigatto *et al* 2022) justified the ineffectiveness of the higher frequency protocols compared to lower frequencies (i.e. for muscle growth) by the similar magnitudes of change in total training volumes (i.e. ‘total load lifted’ Corrêa *et al* 2021, Brigatto *et al* 2022) between the protocols (Corrêa *et al* 2021, Brigatto *et al* 2022). Implicit in this hypothesis is the assumption that skeletal muscle growth is driven by progressive overload. There are several instances (discussed below), however, where progression quantified through changes in total training volume and muscle growth adaptations do not always align (e.g. Takarada *et al* 2000, Kataoka *et al* 2022, Kassiano *et al* 2023). We are not suggesting that increases in total training volume (i.e. progression within training) are unnecessary in the context of a long-term training ‘career,’ but instead that it seems unlikely that muscle growth adaptations observed in 6–12 week resistance training studies stem simply from increases in total training volume.

4.1. Increases in total training volume do not always result in skeletal muscle growth

Many are under the impression that progression within training is necessary for adaptation. Indeed, our group has previously suggested that including variation within resistance training programs geared towards maximizing an adaptation (e.g. muscle growth) may not be necessary over short time frames (e.g. 6–8 weeks) (Hammert *et al* 2021), provided some form of progression is employed (e.g. increase in load, repetitions, etc) (Buckner *et al* 2020b). Given that increases in total training volume will likely occur as result of repeatedly exercising to failure (Jessee *et al* 2018, Buckner *et al* 2020a, Plotkin *et al* 2022), it would seem intuitive that progression within training would be reflective of increases in muscle size. In contrast, however, there is evidence to suggest that simply adhering to the principle of progressive overload (i.e. increasing total training volume) may not always produce skeletal muscle growth (Alway *et al* 1992, Blazevich *et al* 2007, Kacin and Strazar 2011, Laurentino *et al* 2012, Nielsen *et al* 2012, Kataoka *et al* 2022). Observations from Alway *et al* (1992) found that highly competitive bodybuilders were able to progressively increase 10RM training loads (10%–15%) throughout a 24 week elbow flexion resistance training study. However, the researchers (Alway

et al 1992) did not observe muscle growth from pre- to post-intervention. These results agree with a recent study (Kataoka *et al* 2022) wherein 6 weeks of calf resistance training resulted in marked increases in the absolute loads lifted (8–12 and 15–30RMs) without any changes in calf muscle size compared to pre-intervention. Blazevich *et al* (2007) conducted a 5 week study on non-resistance trained individuals, and reported that 4–5 sets of progressive isokinetic knee extension training (6 maximal repetitions per set) produced no greater changes in muscle thickness compared to a time-matched non-exercise control group. In another study on non-trained individuals, lean body mass was shown to increase following three months of progressive resistance training, after which no additional changes in lean body mass were documented in the subsequent six months of progressive training (Volek *et al* 2013). If we assume that the aforementioned training protocols employed were capable of inducing muscle growth (i.e. triggering the mechanotransduction stimulus across the targeted muscle) throughout the entirety of the training period, then these findings (Alway *et al* 1992, Blazevich *et al* 2007, Volek *et al* 2013, Kataoka *et al* 2022) might be suggestive of a noncausal connection between increases total training volume (i.e. progressive overload) and increases in skeletal muscle size.

4.2. Similar magnitudes of progression in total training volume yet different skeletal muscle growth

Blood flow restriction studies have inadvertently demonstrated that similar changes in total training volume may not always produce similar increases in muscle size (Takarada *et al* 2000, Kacin and Strazar 2011, Laurentino *et al* 2012, Nielsen *et al* 2012). For example, Takarada *et al* 2000 had one training protocol exercise to task failure (30%–50% 1RM) with blood flow restriction, and then instructed another to complete the same number of repetitions (also 30%–50% 1RM) without blood flow restriction. Although the two protocols observed similar total training volumes, increases in muscle size were found to favor the blood flow restricted training protocol (Takarada *et al* 2000). The implication of these findings is that ‘similar progression’ in total training volume may not always result in ‘similar stimuli’ for muscle growth. It has also been shown that exclusively performing the initial range of motion during calf resistance training enhanced muscle growth adaptations compared to exercising only in the final range of motion (Kassiano *et al* 2023). The analyses of total training volume (i.e. ‘volume load’ (Kassiano *et al* 2023) included therein revealed that the two training protocols progressed similarly across the 8 week training period (Kassiano *et al* 2023). These results are comparable to Maeo *et al* (2023), who found that overhead position elbow extension training resulted in greater triceps brachii muscle growth compared to training in a neutral arm position, even though the two protocols observed largely similar increases in relative training loads from the initial to final training weeks (Maeo *et al* 2023). If it were true that increases in muscle size could be explained by similar progression in total training volume (i.e. from the initial to final weeks of training) (Damas *et al* 2019, Corrêa *et al* 2021, Brigatto *et al* 2022, Krause Neto *et al* 2022), then it would be difficult to reconcile the results of the abovementioned reports (Takarada *et al* 2000, Kassiano *et al* 2023, Maeo *et al* 2023). It seems more probable that the differential magnitudes of growth observed therein (Takarada *et al* 2000, Kassiano *et al* 2023, Maeo *et al* 2023) were indirectly driven by other extrinsic variables (e.g. proximity to failure Takarada *et al* 2000, range of motion Kassiano *et al* 2023, exercise selection Maeo *et al* 2023). For example, because the addition of blood flow restriction reduces repetitions to task failure compared to non-restricted exercise (using the same relative training load(s)) (Loenneke *et al* 2012), a similar or even lower change in total training volume would still result in a greater stimulus applied to the muscle exercising under restriction. For that reason, the progression in total training volume may simply be a means through which researchers can mathematically detail the total amount of ‘work’ performed over time. This is supported by the results of Lasevicius *et al* (2022), which suggested that simply increasing volume may not augment results of low load training when the level of effort (i.e. proximity to failure) is low.

4.3. Different magnitudes of progression in total training volume yet similar skeletal muscle growth

The aforementioned study by Kassiano *et al* (2023) showed that explicitly performing the final range of motion allowed for greater increases in total training volume (i.e. volume load progression) compared to exercising with a full range of motion. However, increases in muscle size were not different between the two training protocols (Kassiano *et al* 2023), and the initial range of motion protocol demonstrated greater progression in total training volume and increases in muscle size (1 of 2 measured sites) compared to the full range of motion protocol (Kassiano *et al* 2023). In other words, the initial and final range of motion training protocols elicited similar increases in total training volume compared to one another, yet different muscle growth when each was compared to the full range of motion protocol (Kassiano *et al* 2023). A comparison between high and low training protocols (i.e. same exercises and movement patterns) (Schoenfeld *et al* 2015, 2016) found similar muscle growth adaptations (Schoenfeld *et al* 2015), even though the lower load protocol elicited more robust increases in total training volume (i.e. ‘volume load’ (Schoenfeld *et al* 2016)) from the initial to final training week (Schoenfeld *et al* 2016). Low load blood flow restricted training taken

close to task failure (1–4 repetitions away from failure) has also been shown to allow significantly greater increases in ‘volume load’ compared to a performing high load training protocol (1–2 repetitions away from failure), but no greater increases in muscle size after 9 weeks of training (Davids *et al* 2021). It could be that the comparison was limited by the use of two different loading strategies. In fact, the authors (Schoenfeld *et al* 2016) of the former study pointed out that the novelty of the low load training protocol may have provided a greater opportunity to increase total training volume compared to the high load protocol. In addition, total training volume calculations rely heavily on the absolute load lifted and total number of repetitions performed (Dankel *et al* 2017b), which may have influenced increases in total training volume. Data from our laboratory (Jessee *et al* 2018) illustrate similar increases in muscle size between three very low load training protocols and a high load training protocol. The resistance training protocols were all prescribed to task failure or 90 repetitions (whichever occurred first) and were progressed only through increases in the number of repetitions completed (i.e. training loads were not manipulated across sets or throughout the training period) (Jessee *et al* 2018). The average weekly training volumes completed throughout the experiment demonstrate that not all of the very low load training protocols experienced similar magnitudes of training volume progression from the initial to the final week (i.e. 8th week) of training (Jessee *et al* 2018). Yet, even so, similar muscle growth adaptations were observed. The inherent limitation of comparing total training volumes between higher and lower loads would not fully apply here, given that three very low load training protocols all exercised at 15% 1RM (Jessee *et al* 2018). A possible solution proposed by some (Scott *et al* 2016) to rectify differences between higher and lower training loads might be to multiply the number of repetitions performed by the percentage of pre-intervention 1RM rather than absolute training load (i.e. make the training volume measure relative to the specific training protocol). We emphasize, however, that skeletal muscle growth, as a process, is multifactorial (Marcotte *et al* 2015, Lim *et al* 2022), and that the ability to exercise more and consequently amass larger increases in total training volume over time may not inherently produce more muscle growth (Schoenfeld *et al* 2015, 2016, Jessee *et al* 2018, Kassiano *et al* 2023). As mentioned above, quantifying total training volume is method of estimating external variables (i.e. the total amount of ‘work’ performed), not internal variables (i.e. biological processes that underpin skeletal muscle growth; e.g. mechanotransduction).

4.4. Different magnitudes of progression in total training volume and different skeletal muscle growth

While the majority of the current manuscript has focused on scenarios where progression in total training volume and increases in muscle size do not fully align, it seems worthwhile to discuss literature in which they do, and what might explain these results. Four studies (Mattocks *et al* 2017, Dankel *et al* 2017a, 2020, Buckner *et al* 2021) documented more robust muscle growth and increases in total training volume (i.e. from the initial to final weeks of training) for high load training protocols (i.e. 4 sets of 8–12RMs) compared to protocols that performed up to five heavy, single repetitions per session. Given the simultaneous nature of these observations, it might be assumed that the larger progression in total training volume (for the high load protocols) was causative of the greater muscle growth. Although possible, it seems more likely that the ineffectiveness of the heavy single training protocols (i.e. for muscle growth) stems from the fact that they were designed to induce minimal muscle growth (Mattocks *et al* 2017, Dankel *et al* 2017a, 2020, Buckner *et al* 2021). It may be that just one repetition per set was not capable of providing a sufficient amount of contraction-induced time under tension necessary to trigger the mechanotransduction cascade (Dankel *et al* 2017c). If true, then it would seem reasonable that the differences in the amount of exercise volume (i.e. per set) could explain why the two training protocols elicited different increases in muscle size. Though these studies (Mattocks *et al* 2017, Dankel *et al* 2017a, 2020, Buckner *et al* 2021) indicate that different magnitudes of progression in total training volume can result in different muscle growth, it should be noted that it is unclear what muscle growth would have looked like had the high load protocols not been progressed (i.e. in the absence of increases in total training volume). Therefore, it is difficult to conclude whether such differences in muscle growth were truly due to the different magnitudes of progression in total training volume, or the design of each respective protocol.

5. Is progressive overload necessary to induce skeletal muscle growth in research studies?

Two recent studies (Plotkin *et al* 2022, Chaves *et al* 2024) investigated the role of different progression schemes on muscle growth. Neither included a training protocol that was not progressed, which might stem from a number of reasons, but nonetheless highlights the common belief that progressive overload is necessary for muscle growth. The American College of Sports Medicine guidelines for resistance training (Ratamess *et al* 2009) recommend that there should be an increase in the demands placed on the musculature (i.e. progressive overload) to allow the possibility for additional increases in muscle size.

However, these statements appear to be rooted in intuition, or experience, as there are no citations provided. While it is logical to think that progressive overload would be important throughout a training 'career,' it remains plausible that in resistance training studies and/or over shorter time frames (e.g. 6–12 weeks) an approach to training which does not include progressive overload could be similarly as effective for stimulating muscle growth compared to one that does.

Twice daily bench press training over a two week time frame (i.e. 6 d per week; 24 total training sessions) with blood flow restriction has been shown to increase muscle size compared to the same training without restriction (Yasuda *et al* 2010). Training load and volume, respectively, were set at 30% 1RM and 75 total repetitions (i.e. $30 + 3 \times 15$ repetitions) and remained constant throughout the study (Yasuda *et al* 2010). Kubo *et al* (2006) used a similar approach and found changes in muscle size with blood flow restriction comparable to that of a condition trained with progressively increased high loads (4 set of 10 repetitions using 80% 1RM). However, both studies progressed the blood flow restriction pressure which could have provided some form of overload (Kubo *et al* 2006, Yasuda *et al* 2010). Nonetheless, these findings provide some evidence to suggest that progressively overloading the resistance exercise stimulus in more traditional ways (e.g. increases in load, repetitions) may not always be required for muscle growth. However, the lack of a time matched, non-exercise control group precludes a definitive conclusion on the topic. To the best of our knowledge, no study has demonstrated the importance of (or lack thereof) progressive overload for increasing muscle size.

It remains possible that an exercise stimulus which is capable of inducing skeletal muscle growth at the beginning of a training study, but is not progressed further, would result in similar increases in muscle size compared to a protocol that is progressively overloaded (and thus, increases total training volume). For example, even if regular engagement in resistance exercise induced an adaptation such that less muscle is required to lift the same submaximal load, for the same number of repetitions (Ploutz *et al* 1994), it is plausible the mechanotransduction cascade may still occur across a large proportion of the targeted muscle. If the mechanotransduction cascade still occurs across a large proportion of the targeted muscle in the absence of progressive overload, then the protocol would likely remain effective for increasing muscle size. Indirect support for such a statement was shown by Brigatto *et al* (2019), who documented similar muscle growth between training protocols that exercised twice and four times per week, even though the twice weekly protocol was unable to increase the total load lifted from the first to eight training week (Brigatto *et al* 2019). These results should be interpreted cautiously, however, because the study was not explicitly designed to address the importance of progressive overload for muscle growth, and the total load lifted was summed across nine different exercises (Brigatto *et al* 2019). In addition, it is worth mentioning that, at some point, it seems likely an exercise stimulus which is not progressed would become incapable of stimulating the mechanotransduction cascade across a large proportion of the targeted muscle. The point at which further overload becomes necessary to provide an effective stimulus for additional increases in skeletal muscle size likely depends on a number of different factors (e.g. total number of sets performed per training session), and currently remains speculative as little work has been conducted on the topic.

6. Progressive overload in resistance training studies—future directions?

From a research standpoint, it seems worthwhile to address the importance of progressive overload for increasing muscle size, as it may help better understand skeletal muscle growth adaptations to resistance training. In theory, researchers could test such in many ways, albeit for simplicity we would suggest that changes in muscle size be compared between one resistance training protocol that has progressive overload applied (and thus, increases total training volume) and another that is not progressively 'overloaded' beyond the initial training visit (and thus, does not increase total training volume). For example, the two training protocols could perform each set of exercise to task failure on the initial visit (e.g. 4 sets to failure at 70% 1RM). The first protocol (i.e. 'overloaded' protocol) would then continue to exercise to failure (i.e. each training session) at the same percentage of maximal strength (i.e. 70% 1RM), which would likely result in progressive increases in repetitions and thus, total training volume (Jessee *et al* 2018, Buckner *et al* 2020a). Note that investigators may also induce progressive overload through increasing the absolute load lifted per repetition (e.g. 8–12RM) (Plotkin *et al* 2022, Hammert *et al* 2023b). The second protocol (i.e. 'nonoverloaded' protocol) would complete the same exact exercise which was completed on the initial training visit (to avoid progression). We also wish to suggest that a time-matched, non-exercise control group be used to capture random error across time (Dankel *et al* 2020). It is our opinion that performing additional repetitions and/or sets of exercise, increasing the absolute load, etc in attempt to chase a mathematically derived total training volume may be less important in terms of stimulating skeletal muscle growth than what has been recently suggested (Damas *et al* 2019, Corrêa *et al* 2021, Brigatto *et al* 2022, Krause Neto *et al* 2022). In fact, we speculate that the results of the aforementioned design would show no

hypertrophic advantage of progressively overloading the resistance exercise stimulus after 6–8 weeks of training. If true, then such findings may open doors for future study designs to examine the efficacy of different resistance training protocols for increasing skeletal muscle size. We remind the reader that we are not suggesting that progressive overload, as detailed by increases in total training volume over time, is not important for muscle growth throughout a training ‘career.’ Moreover, we wish to make it clear that the focal point of the current discussion was on the influence of increases in total training volume from the initial to final week of training for increasing muscle size. Whether different tissues (e.g. bone, tendon) and/or other adaptations to resistance training (e.g. increases in muscular strength or local endurance) would benefit from progressive overload is beyond the scope of this paper, but could be an area of future study.

7. Conclusion

While there is evidence to support a physiological rationale for including progressive overload within resistance training studies (Ploutz *et al* 1994, Phillips *et al* 2002), and reason to think that it would be important throughout a long-term training ‘career,’ we wish to suggest quantifying increases in total training volume (i.e. as a surrogate of progressive overload within research studies) may be superfluous and inform little on muscle growth adaptations. Our suggestion is based on the following research findings: (1) a noncausal connection between increases in total training volume (i.e. progressively overloading the resistance exercise stimulus) and increases in skeletal muscle size (Alway *et al* 1992, Volek *et al* 2013, Kataoka *et al* 2022); (2) similar changes in total training volume may not always produce equivalent increases in muscle size (Takarada *et al* 2000, Nielsen *et al* 2012, Kassiano *et al* 2023, Maeo *et al* 2023); and (3) the ability to exercise more and amass larger increases in total training volume from the initial to final week of training may not inherently produce more skeletal muscle growth (Schoenfeld *et al* 2015, 2016, Jessee *et al* 2018, Kassiano *et al* 2023). It is our position that performing additional repetitions and/or sets of exercise, increasing the absolute load, etc in an attempt to chase a mathematically derived total training volume may be less important in terms of stimulating skeletal muscle growth compared to what has been suggested (Damas *et al* 2019, Corrêa *et al* 2021, Brigatto *et al* 2022, Krause Neto *et al* 2022). Quantifying changes in total training volume may simply be a means through which researchers can mathematically determine the total amount of external ‘work’ performed within a resistance training study.

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Author contributions

WBH, RK, YY, JSS, AK, RWS, and JPL discussed concerns with current literature and provided initial discussion for a narrative review. WBH wrote the initial draft of the manuscript. RK, YY, JSS, AK, RWS, and JPL provided individual critiques of the manuscript. WBH revised manuscript accordingly. All authors approved final version

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